KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

HOSPITAL CHATBOT FOR COLOR-CODED CLINICAL PATHWAYS

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DECLARATION

We hereby declare that this submission is our own work towards the B.Sc. degree and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the University, except where due acknowledgement has been made in the text.

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DEDICATION

This work is dedicated to patients and frontline triage staff in Ghanaian emergency departments, whose safety depends on timely, colour-coded clinical pathways.

ABSTRACT

Emergency departments in low- and middle-income countries (LMICs) face overcrowding, delayed first assessment, and a failure we call *acuity discontinuity*: the South African Triage Scale (SATS) colour assigned at the desk often stops guiding care after the patient leaves triage. This thesis presents a hospital chatbot that maps an unstructured chief complaint to a SATS acuity level and colour, then attaches a concrete clinical pathway—destination, target response time, and actions.

Part 1 implements a Digital Acuity Engine: a medical gate filters non-clinical text; optional OpenMed named-entity enrichment surfaces diseases and drugs; a fine-tuned BioBERT classifier predicts acuity levels 1–5; a fixed map f_{SATS} yields Red, Orange, Yellow, or Green. On an 8,000-sample stratified holdout, baseline BioBERT achieved 99.92% colour-routing accuracy and 98.14% Level-1 (Red) recall.

Part 2 defines protocol-mapped pathways $P(C)$ for each colour (including Green diversion from the acute ED and a Blue dignity protocol). The architecture deliberately omits Bayesian fusion with Triage Early Warning Score (TEWS) vitals at chatbot intake: BioBERT already performs end-to-end text-to-acuity inference; pathways attach deterministically to colour. TEWS remains nurse-side or future work.

The contribution is a defensible LMIC chatbot design and prototype (Curatio) that answers whether NLP can drive colour-coded clinical pathways without requiring a full TEWS–Bayesian stack before the triage desk.

Keywords: SATS; clinical pathways; BioBERT; triage chatbot; LMIC; acuity discontinuity.

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CONTENTS

ABBREVIATIONS

SATS South African Triage Scale

TEWS Triage Early Warning Score

ED Emergency Department

OPD Outpatient Department

LMIC Low- and Middle-Income Country

KATH Komfo Anokye Teaching Hospital

NLP Natural Language Processing

NER Named Entity Recognition

BioBERT Biomedical Bidirectional Encoder Representations from Transformers

SMOTE Synthetic Minority Over-sampling Technique

PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses

API Application Programming Interface

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 Background of the study

In high-functioning emergency care, **triage** prioritises patients by urgency so that the right patient receives the right care at the right time. The **South African Triage Scale (SATS)**—widely used in LMIC settings including Ghana—assigns colour codes: Red (immediate), Orange (very urgent, typically within 10 minutes), Yellow (urgent, typically within 60 minutes), Green (non-urgent), and Blue (deceased / dignity pathway).

These colours are meant to trigger **clinical pathways**: standardised roadmaps of destination, diagnostics, and time targets. In practice, however, colour often becomes a label that ends at the triage desk. Once a patient moves to imaging, a ward, or a generic queue, the hospital ecosystem may "forget" their acuity. We call this failure **acuity discontinuity**.

Compounding the problem is the **Green Tag Burden**: non-urgent patients occupy acute emergency capacity. Pre-triage queues may also delay first assessment for hours. Patients arriving via digital channels can describe symptoms in free text long before six TEWS vitals are available.

This project develops a **hospital chatbot for color-coded clinical pathways**. The chatbot collects a chief complaint, predicts SATS acuity and colour with clinical NLP, and attaches an operational pathway card—without requiring a TEWS–Bayesian fusion engine at intake.

1.2 Problem statement

Resource-constrained hospitals face:

1. **Pre-triage bottleneck** — delay before any colour is assigned.
2. **Cognitive load** — manual TEWS calculation under pressure risks under- and over-triage.
3. **Static triage** — one-shot colour without digital timers or re-assessment prompts.
4. **Lost urgency** — Orange/Yellow protections are not tracked after the ED desk.
5. **Text-only intake** — vitals are often missing at home or in the queue, so a language-based acuity signal is needed at the door.

Generic consumer symptom checkers are not aligned to SATS hospital pathways or local operational routing (e.g. Polyclinic vs acute ED).

1.3 Aim and objectives

1.3.1 General objective

To design and prototype a chatbot-mediated Digital Acuity Engine that maps unstructured chief complaints to SATS colour-coded clinical pathways for LMIC hospital intake.

1.3.2 Specific objectives

1. Fine-tune and evaluate a BioBERT classifier that maps chief complaints to acuity levels 1–5.

2. Map acuity to SATS colour via a fixed function f_{SATS} and report colour-routing performance.
3. Define protocol-mapped pathways $P(C)$ for Red, Orange, Yellow, Green, and Blue.
4. Implement a chatbot/API prototype (Curatio) with medical gating and optional clinical NER enrichment.
5. Document a defensible architecture that uses NLP acuity **without** Bayesian TEWS fusion at chatbot intake.

1.4 Research questions

RQ1 Can unstructured chief complaints be mapped to SATS acuity levels 1–5 with clinically usable accuracy?

RQ2 Does correct acuity prediction imply correct colour assignment under fixed f_{SATS}?

RQ3 How should colours translate into pathways (destination, $T_{\max}$, actions)?

RQ4 Can a chatbot deliver pre-triage acuity and pathway guidance before the nurse desk?

RQ5 How well does the system reject non-clinical input?

RQ6 Does OpenMed NER enrichment improve interpretability without replacing the acuity model?

RQ7 How does class imbalance affect Level-1 (Red) recall, and what mitigation is appropriate?

RQ8 How overconfident is softmax output, and what does that imply for nurse oversight?

RQ9 Can Green-pathway diversion be specified to reduce acute ED congestion?

RQ10 What LMIC architecture uses NLP acuity for colour pathways without a TEWS–Bayesian stack at intake?

1.5 Scope and limitations

In scope: text → BioBERT → acuity → colour → pathway protocols; medical gate; OpenMed enrichment; holdout evaluation; Curatio prototype.

Out of scope: Bayesian networks; three-layer fusion of NLP+TEWS+Bayes; live TEWS calculator as a chatbot requirement; prospective KATH clinical trial; full HIS integration.

TEWS and Bayesian methods appear in related literature and older project slides; this thesis treats them as **future work / nurse-desk overlays**, not Part 2 deliverables.

1.6 Contributions

- An implemented NLP Digital Acuity Engine with measured holdout performance.
- A protocol table linking SATS colours to hospital pathways (including Green diversion).
- A clear architectural claim: deterministic colour and pathway attachment after end-to-end text classification, without Bayesian fusion at intake.
- Open framing documents and a first-draft thesis structure for continued writing.

1.7 Organisation of the thesis

Chapter 2 reviews SATS, overcrowding, and triage NLP. Chapter 3 presents methodology. Chapter 4 reports results and discusses pathways. Chapter 5 concludes and states future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter situates the project in emergency triage practice (SATS), LMIC overcrowding, clinical NLP for triage, and conversational systems. Systematic-review reporting can follow PRISMA 2020 templates archived under `systematic-review-prisma/`. Detailed paper notes belong in `research-findings/`.

2.2 Emergency overcrowding and the Green Tag Burden

Emergency departments worldwide face overcrowding. A structural driver is the influx of non-urgent patients into acute facilities—the so-called Green Tag Burden. In LMICs, understaffing and limited diagnostics worsen the competition for beds and clinician time. Digital diversion of Green patients to OPD, Minors, or Polyclinic streams is therefore not only a convenience feature but a flow-management requirement for colour-coded pathways.

Waiting time interacts with colour assignment and outcomes; literature on waiting time and triage colour codes informs why pathway timers ($T_{\max}$) matter after a colour is assigned.

2.3 SATS in LMIC and Ghanaian practice

SATS combines physiological early warning (TEWS) with clinical discriminators that can override mild vital-sign scores when high-risk symptoms are present (e.g. central chest pain). Implementation studies at Komfo Anokye Teaching Hospital (KATH) and in other LMIC ambulance/ED settings demonstrate that SATS is operationally relevant beyond South Africa.

Implication for this thesis: Colour codes already exist in hospital policy. The research gap is not inventing colours, but (i) assigning them from free-text complaints before full vitals, and (ii) binding colours to pathway protocols so acuity does not disappear after the desk.

2.4 Triage Early Warning Score (TEWS) — context only

TEWS aggregates mobility, respiratory rate, heart rate, temperature, AVPU, and trauma into a sum that maps to colour bands. It is central to bedside SATS. However, chatbot intake often lacks complete vitals. This thesis therefore does **not** implement TEWS algebra in the chatbot. TEWS remains the nurse-side gold-standard overlay after arrival.

2.5 Why Bayesian fusion is deferred

Multimodal Bayesian triage models appear in the literature as ways to combine uncertain evidence. Older project materials sketched $P(C \mid E)$ fusion with TEWS and NLP. For this thesis we defer Bayesian fusion because:

- BioBERT already maps text to a full acuity distribution via softmax;
- duplicating a second probabilistic layer at intake expands scope without matching the deployed prototype;

- deterministic f_{SATS} and $P(C)$ already produce actionable colour pathways.

Bayesian papers are retained as related work (skim) and future research, not methodology requirements.

2.6 Clinical NLP and BioBERT

Transformer models—BERT and domain-adapted BioBERT—enable contextual embeddings for biomedical text. Deep attention models have been applied to ED triage notes. Reviews of NLP at ED triage (e.g. Stewart and colleagues) summarise opportunities and risks. Classification with a fixed label set reduces open-ended generative hallucination risk compared with unconstrained large language models.

Our pipeline fine-tunes BioBERT for five-class acuity prediction on chief complaints, optionally enriched with OpenMed NER tags for diseases and medications.

2.7 Conversational agents and digital triage tools

Symptom-assessment applications and ED prediction chatbots show that conversational NLP can support triage-like advice. Validation studies emphasise safety, under-triage monitoring, and comparison with conventional triage. Our chatbot is constrained: it outputs acuity, colour, and pathway cards aligned to SATS—not free-form diagnosis essays.

2.8 From colour labels to clinical pathways

Clinical pathways are step-by-step care roadmaps. Synopsis and clinical protocol materials in this project define Five-Path logic: Red resuscitation, Orange countdown, Yellow parallel standing orders, Green diversion, Blue dignity. The literature supports treating colour as operational, not decorative.

2.9 Summary of the gap

Prior art provides SATS colours, TEWS bedside scoring, and growing NLP triage evidence. Missing for LMIC chatbot intake is an integrated story: accurate text→acuity→colour **plus** explicit pathway protocols, without mandating Bayesian TEWS fusion before the nurse has vitals. Chapters 3–4 address that gap.

CHAPTER 3

METHODOLOGY

3.1 Overview

The methodology has two layers:

1. **Digital Acuity Engine (Part 1):** text $\to$ acuity $\hat{c} \to$ colour $C = f_{\text{SATS}}(\hat{c})$.
2. **Pathway protocols (Part 2):** $C \mapsto P(C) = (T_{\max}, \text{dest}, \text{actions}, \text{escalation})$.

No Bayesian fusion or live TEWS calculator is included in the chatbot inference path.

3.2 Data and fine-tuning

Chief-complaint text paired with acuity labels was used to fine-tune BioBERT for sequence classification over five classes. Training artefacts and notebooks live under `fine-tuned-biobert/`. A stratified 10% holdout (n = 8,000) was reserved for evaluation. An optional SMOTE-augmented training variant was compared for minority-class (Red) recall.

3.3 Medical gate

Before acuity prediction, a clinical relevance gate scores whether the input is a medical chief complaint. Non-clinical text (greetings, shopping, sports, etc.) is rejected with a dynamic message and category (e.g. `non_clinical_topic`). Default threshold: 0.35. This protects the triage model from nonsense inputs and answers RQ5.

3.4 OpenMed enrichment

When enabled, OpenMed NER extracts disease/condition and medication spans, with simple negation heuristics. Optional entity prefixes of the form `[DISEASE: ...]` / `[DRUG: ...]` may be prepended before BioBERT tokenisation. OpenMed does **not** predict acuity; BioBERT remains the classifier (RQ6).

3.5 BioBERT inference

Let X be the (possibly enriched) complaint. BioBERT produces logits $\mathbf{z} \in \mathbb{R}^5$. Softmax yields

$$\hat{y}_c = \frac{e^{z_c}}{\sum_{j=1}^{5} e^{z_j}}, \qquad \hat{c} = \arg\max_c \hat{y}_c, \qquad \text{confidence} = \max_c \hat{y}_c.$$

Training minimises cross-entropy against nurse acuity labels.

3.6 Colour mapping

$$C = f_{\text{SATS}}(\hat{c}) = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases}$$

Implemented in the ML service as a constant lookup. Under this fixed map, colour-routing accuracy equals acuity accuracy (RQ2).

3.7 Pathway protocols $P(C)$

Blue is a dignity protocol outside the five-class acuity head. The chatbot should emit a pathway card alongside colour and confidence.

Table 3.1: Protocol-mapped colour pathways (chatbot Part 2).

Colour	$T_{\max}$	Destination	Actions (summary)
Red	0 min	Resuscitation	Code Red; bypass registration
Orange	10 min	Acute bed	Countdown; escalate if unclaimed
Yellow	60 min	Urgent ED stream	Standing orders in parallel
Green	Routine	OPD / Minors / Polyclinic	Divert from acute ED
Blue	Dignity	Mortuary pathway	Silent social-work protocol

3.8 Evaluation protocol

On the holdout set we report accuracy, precision, recall, F1, macro-F1, per-class support, confusion matrices, mean confidence, and calibration plots. Level-1 recall is emphasised as a safety metric (RQ7). Softmax overconfidence informs nurse-in-the-loop guidance (RQ8).

3.9 System prototype (Curatio)

- **ML service:** FastAPI — `/predict`, `/analyze`, `/deidentify`, voice intake.
- **API gateway:** Express proxies to the ML service.
- **Client:** React triage test page and presentation dashboard.

Default predict path enables the medical gate and OpenMed enrichment; both can be disabled for ablation demos.

3.10 What is not in the methodology

TEWS piecewise scoring, Bayesian posteriors $P(C_k \mid E)$, and fusion max-urgency controllers are excluded from the implemented method. They may appear in future work after vitals capture at the desk.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Holdout classification performance

Evaluation used a stratified 10% holdout of n = 8,000 complaints. Summary metrics:

Table 4.1: Baseline vs SMOTE BioBERT on stratified holdout.

Metric	Baseline	SMOTE
Accuracy	0.9992	0.9985
Macro-F1	0.9977	0.9954
Recall L1 (Red)	0.9814	1.0000
Mean confidence	0.9997	0.9992
% confidence = 1.0	79.05%	82.99%
Latency p50 (ms/sample)	100.3	95.5

RQ1: Baseline BioBERT maps complaints to acuity with very high holdout accuracy. **RQ2:** With fixed f_{SATS}, colour-routing accuracy matches acuity accuracy (7,994/8,000 = 0.9992).

Default inference uses the baseline checkpoint (slightly higher overall accuracy/macro-F1). SMOTE maximises Red recall on this holdout at a small cost to overall metrics—a clinical trade-off for RQ7.

4.2 Errors and Red safety

All six baseline errors on the reported holdout involved true Level 1 (Red) cases routed to a less urgent colour ($TP = 316$, $FN = 6$, $FP = 0$ for L1). This underscores why Red recall and nurse oversight matter even when overall accuracy is high.

4.3 Calibration and confidence

Mean confidence is extremely high; about 79% of baseline predictions have confidence exactly 1.0. Calibration plots show overconfidence. **RQ8:** pathway assignment should remain advisory to clinicians; low-confidence or safety-critical presentations warrant bedside TEWS and human confirmation. Overconfidence is **not** solved by adding Bayesian fusion in this thesis; it motivates process controls.

4.4 Medical gate and OpenMed (qualitative / system tests)

Non-clinical prompts (e.g. sports, streaming) are rejected with explanatory messages when the gate is enabled (RQ5). OpenMed enrichment surfaces disease and drug spans (e.g. chest pain, dyspnoea, metformin) alongside acuity, improving interpretability without replacing BioBERT (RQ6).

4.5 Pathway scenarios (Part 2 discussion)

Although the production API currently returns colour labels (and related fields), Part 2 defines operational meaning:

- **Orange chest-pain scenario:** language suggestive of ACS $\rightarrow$ Level 2/Orange $\rightarrow$ acute bed + 10-minute countdown pathway card.
- **Green minor complaint:** Level 4/5 $\rightarrow$ Green $\rightarrow$ divert to OPD/Polyclinic (RQ9), addressing Green Tag Burden.
- **Red immediate:** Level 1 $\rightarrow$ Code Red / resus pathway with $T_{\max} = 0$.

These scenarios show how colour becomes a pathway without Bayesian vitals fusion (RQ3, RQ10).

4.6 Limitations

- Holdout is in-distribution relative to training complaints; live KATH language may differ.
- Softmax overconfidence limits trustworthy autonomous routing.
- Pathway timers and HIS escalation are designed, not fully instrumented as hospital middleware.
- TEWS bedside scoring is not automated in the chatbot.
- No prospective clinical trial is claimed in this draft.

4.7 Discussion relative to hybrid TEWS–Bayes designs

Hybrid designs that fuse TEWS, Bayes, and NLP are intellectually attractive when vitals exist. At pre-desk chatbot intake they are often under-determined. Our results support a staged architecture: NLP colour pathways first; TEWS when measured; richer probabilistic fusion only if prospectively justified later.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This thesis addressed **Hospital Chatbot for Color-Coded Clinical Pathways** in two parts. Part 1 delivered a working Digital Acuity Engine: medical gating, optional clinical NER, fine-tuned BioBERT acuity prediction, and deterministic SATS colour mapping, with strong holdout colour-routing performance. Part 2 specified protocol-mapped pathways so each colour carries destination, time target, and actions—including Green diversion and Blue dignity—thereby attacking acuity discontinuity without pretending that a badge alone is a pathway.

We explicitly **did not** implement Bayesian fusion. Softmax already yields an acuity distribution from text; pathways attach by f_{SATS} and $P(C)$. That answers RQ10 with a lean LMIC intake architecture.

5.2 Answers to research questions (summary)

- **RQ1–RQ2:** Yes—high holdout acuity/colour accuracy under fixed f_{SATS}.
- **RQ3, RQ9:** Pathways defined as protocol table including Green diversion.
- **RQ4:** Curatio chatbot/API demonstrates pre-desk delivery of acuity and colour (pathway card as design completion).
- **RQ5–RQ6:** Gate and OpenMed behave as intended in system tests.
- **RQ7–RQ8:** Red recall and calibration overconfidence require safety-aware deployment.

- **RQ10:** NLP$\rightarrow$colour$\rightarrow P(C)$ is a defensible intake stack without TEWS–Bayes.

5.3 Contributions restated

1. Implemented and evaluated text-to-SATS-colour triage NLP.
2. Formalised colour-coded clinical pathway protocols for chatbot output.
3. Scoped mathematics honestly: softmax + deterministic maps in; Bayesian fusion out.

5.4 Future work

- Instrument pathway cards and timers in the live API/UI.
- Optional nurse-desk TEWS calculator after vitals capture (not Bayesian-by-default).
- Prospective evaluation with KATH clinicians; measure under-triage and time-to-provider.
- Domain adaptation to Ghanaian informal English / Twi-translated complaints.
- Only if justified: compare Bayesian multimodal fusion as a *post-vitals* research extension—not a rewrite of Part 1.

REFERENCES

APPENDIX

5.5 Formal maps

Acuity to colour:

$$C = f_{\text{SATS}}(\hat{c}).$$

Pathway:

$$P(C) = \big(T_{\max}(C),\, \text{dest}(C),\, \text{actions}(C),\, \text{escalation}(C)\big).$$

5.6 Repository pointers

- Prototype: `curatio/`
- Model training: `fine-tuned-biobert/`
- Metrics: `results/eval_outputs/`
- Framing: `project-framing/`
- Reading notes: `research-findings/`
- PRISMA templates: `systematic-review-prisma/`

5.7 Evaluation artefacts

Confusion matrices and calibration plots: `results/eval_outputs/baseline_*.png`, `smote_*.png`.